Introduction to Causality

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Outline

1. Overview
2. Bayesian Network
3. Causal Bayesian Network
4. Structural Causal Model

1. Overview

- **What is causality?**
  In the normal semantics of machine learning, causality refers to understanding and modeling cause-and-effect relationships.
- **Causality and Correlation are different or not?**

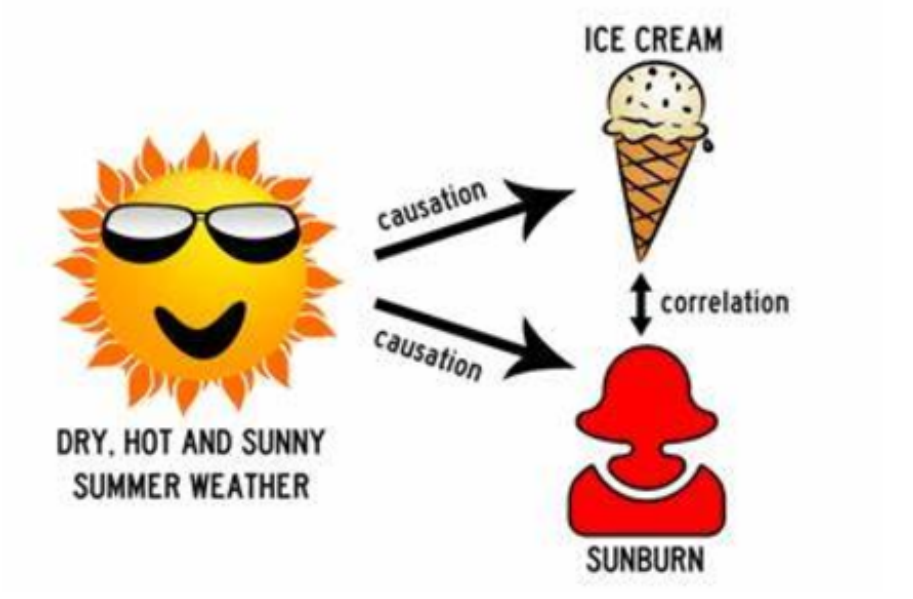

Figure: Causal and Correlation

Causality $\rightarrow$ dependence, But dependence $\rightarrow$ causality

Bayesian Network

Definition: A Bayesian network (directed graphic model) is a directed acyclic graph representing a probabilistic model between variables. It represents a joint distribution over variables as a product of local conditional distributions.

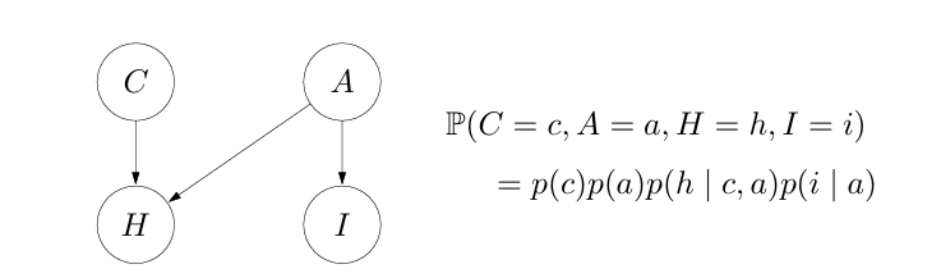

Figure: Bayesian Network

$$P(X_1 = x_1, \ldots, X_n = x_n) = \prod_{i=1}^{n} P(X_i \mid PA_{X_i})$$

2. Bayesian Network

Normal Concept

- **D-separation** In DAG, two nodes X and Y are d-separated by a set of nodes Z if every path between X and Y is blocked given Z.
  denote: $sep_G(X, Y|Z)$
- **I-map** G is an I-map for P if every conditional independence implied by the d-separation in G holds in the distribution P.
- **P-map** A graph G is a perfect map of P if G captures all and only the conditional independencies in P.

2. Bayesian Network

Markov property

Given a DAG G and a join distribution P^X this distribution is said to satisfy.

- **Global Markov property** with respect to the DAG G if $sep_G(X, Y|Z) \implies X \perp Y|Z$
- **Local Markov property** with respect to the DAG G if $X \perp non - Descedant_x \perp Parents_x$
- **Markov factorization property** with respect to the DAG G if $p(x) = p(x_1, ..., x_n) = \prod_{j=1}^{n} p(x_j|x_{PA_j})$

2. Bayesian Network

Inference in BN

- Probabilistic inference refers to computing the probabilistic of each variable based on variable observation
- **Type of inference**:
  Exact inference: variable elimination, a junction tree algorithm.
  Approximation inference: variational inference, sampling (Monte Carlo methods, Markov chain Monte Carlo, Gibbs Sampling).

Why Causal Models?

As mentioned, an Inference Bayesian network computes the marginal or conditional dependence distribution of partial variables in observable variables.

$\Longrightarrow$ what happens with noise data?

Consider An Example (Kidney Stone)

	Treatment A	Treatment B
Small Stones ($\frac{357}{700} = 0.51$)	$\frac{81}{87} = 0.93$	$\frac{234}{270} = 0.87$
Large Stones ($\frac{343}{700} = 0.49$)	$\frac{192}{263} = 0.73$	$\frac{55}{80} = 0.69$
	$\frac{273}{350} = 0.78$	$\frac{289}{350} = 0.83$
	$\frac{562}{700} = 0.80$	

Figure: Medician Kidney Stone Example

$\Longrightarrow$ Decision treatment will change based on stone size.

3. Causal Bayesian Network

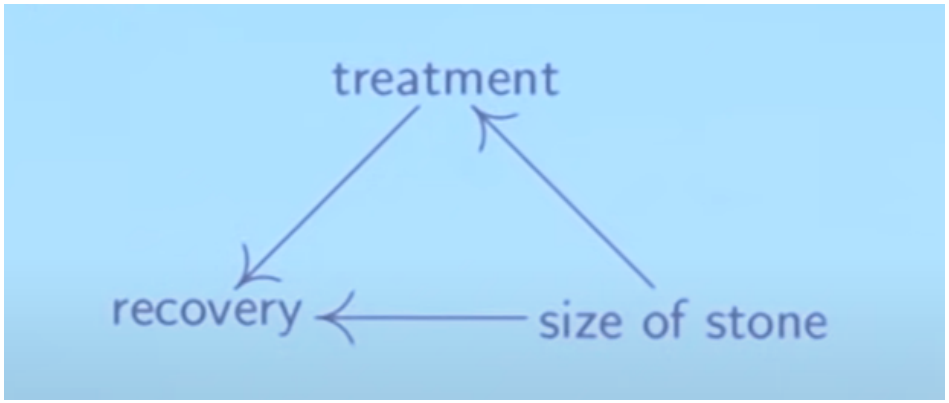

Figure: Kidney Medician Graphic

- Normal question in BN, How would seeing X change my belief in Y? (seeing)

$$P(recovery|treatment)$$

- Second question, what happens to Y if I do X? (doing)

$$P(recovery|do(treatment = B), sizestone)$$

- Third question: Was it x that caused Y? (imagining)

$$P(recovery_{treatment=A}|treatment = B, sizestone)?$$

3. Causal Bayesian Network

$\Longrightarrow$ The main idea in the causal model is that can intervention and counterfactuals

- **Intervention**: refers to what happens to variable A after "do" (intervene) on a variable B.
- **counterfactuals**: refer to hypothetical scenarios or "what-if" analyses that examine how the outcome of a system would change if certain conditions were different from what occurred.

$\Longrightarrow$ which tool is used to model causal?

4. Structural Causal Model

Structural Equation Model (SEM)

Definition: A structural equation model is defined as a tuple $S = (s, P^N)$, where:

- $s = (s_1, \ldots, s_p)$ is a collection of p equations:

$$s_j : X_j = f_j(\mathrm{PA}_j, N_j), \quad j = 1, \ldots, p,$$

  where PA_j represents the parents of X_j.
- P^N is the joint independent distribution of the noise variables.

Example of SEM

$$X_1 = f_1(X_3, N_1)$$
$$X_2 = f_2(X_1, N_2)$$
$$X_3 = f_3(N_3)$$
$$X_4 = f_4(X_2, X_3, N_4)$$

- N_i jointly independent
- $\mathcal{G}_0$ has no cycles

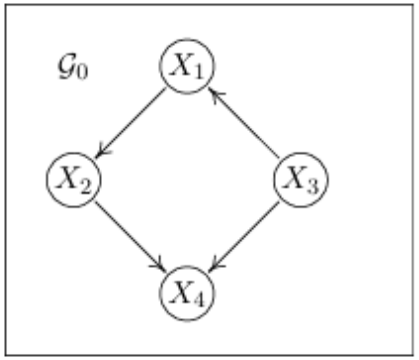

Figure: Simple SEM structure

By only topology pi = X3 $\rightarrow$ X1, X1 $\rightarrow$ X2, X2 , X3 $\rightarrow$ X4 perform distribution observed data and intervention distribution.

4. Structural Causal Model

Intervention Distribution

Definition: Consider a distribution P^X generated from a Structural Equation Model (SEM) $S = (s, P^N)$. An **intervention distribution** is obtained by:

- Replacing one (or more) structural equations in S with new equations (ensuring no cycles in the graph),
- Resulting in a new SEM $\hat{S}$.

The new distribution in $\hat{S}$ is defined as:

$$P^X_{\hat{S}} = P^{X|\mathrm{do}(X_j = f(\hat{\mathrm{PA}}_j, \hat{N}_j))}_S$$

Key cases of intervention:

- $P^X_{\hat{S}} \neq P^X_S$: **Perfect intervention**.
- $P^X_{\hat{S}} = P^X_S$: **Imperfect intervention**.

4. Structural Causal Model

Causal Effect

Definition: Given a Structural Equation Model (SEM) S, there is a total causal effect from X to Y if and only if:

$$X \not\perp\!\!\!\perp Y \in P_S^{X|\mathrm{do}(X=\hat{N}_X)}$$

for some variable $\hat{N}_X$.

Example : Suppose that a linear system
$P^{X,Y}$is induced by a structural equation model S: $X = N_x, Y = 4.X + N_y$
With $N_x, N_y \approx^{iid} N(0,1)$ and graph X $\implies$ Y.

$$P_S^{Y|do(N_X=2)} = N(8,1) \neq N(0,17) = P_S^Y$$

But one other hand:

$$P_S^{X|do(y=2)} = P_S^Y = N(0,1)$$

4. Structural Causal Model

Proposition (SEM)

Given an SEM S, the following statement are quirvalent

- If there is no directed path from X to Y, then there is no causal effect. Otherwise, a directed graph also can have no causal effect.
- There is x and $x^{'}$, such that $P_S^P Y|do(X = x) \neq P_S^{Y|do(X=x')}$
- There is x , such that $P_S^{Y|do(X=x)} \neq P_S^Y$

4. Structural Causal Model

Counterfactual Distribution

Definition: Consider a Structural Equation Model (SEM) $S = (s, P^N)$ over nodes X. Given some observation $X = x$, we define a **counterfactual SEM** by replacing the distribution of noise variables:

$$S_{X=x} = (s, P^N_{X=x}),$$

where $P^N_{X=x} = P^N \mid X = x$.

- In this new SEM, the set of noise variables $P^N_{X=x}$ need not be mutually independent.
- Counterfactual statements can now be interpreted as **do-statements** within the new counterfactual SEM.

4. Structural Causal Model

Example of Counterfactual: Consider the following made-up scenario: A patient with poor eyesight comes to the hospital and goes blind ($B = 1$) after the doctor suggests the treatment ($T = 1$). Let us assume that the correct SEM has the following form:

- $T = N_T$
- $B = T \cdot N_B + (1 - T) \cdot (1 - N_B)$

$$S : T \to B \quad | \quad B \sim \text{Ber}(0.01)$$

A question: "What would have happened had the doctor decided to give treatment $T = 0$?"

We have the factual observation: $T = 1, B = 0$. By counterfactual reasoning

$$P(B = 1|do(T = 1)) = 0.99$$

After counterfactual , $P(B = 0|do(T = 0)) = 0.99$

$\implies$ counterfactual make change of SEM.

4. Structural Causal Model

Causal Property

- **Faithfulness:** P^X is said to be faithful to the DAG G if $Sep_G(A, B|C) \Longleftarrow A \perp B|C$
  $\Longrightarrow$ In other words, no causal effect with an independent variable in the faithful causal model
- **Causal minimality** A distribution satisfies causal minimality concerning G if it is Markov concerning G, but not to any proper subgraph of G.
  $\Longrightarrow$ can not remove the dependent variable in the subgraph with no causal change in G.

Some Other Property

Proposition 4.1 Consider the random vector $X = (x_1, x_2, ..., x_p)$ and assume that the joint distribution has a density concerning a product measure. Suppose that P_X is Markov with respect to G. Then P_X satisfies causal minimality with respect to G if and only if $X_j Y \in PA_j^G$ we have that $X_j \perp Y|(PA_j^G/Y)$.

Proposition 4.2 Assume that P^X is generated by an SEM with graph G. Then, P^X is Markov with respect to G.

Proposition 4.3 Consider $(x_1, x_2, ..., x_p)$ and let P^X have strictly positive and it is Markov with respect to G. Then there exits an SEM (S, P_X) with graph G that generates the distribution P^X,

Overview of Causal model

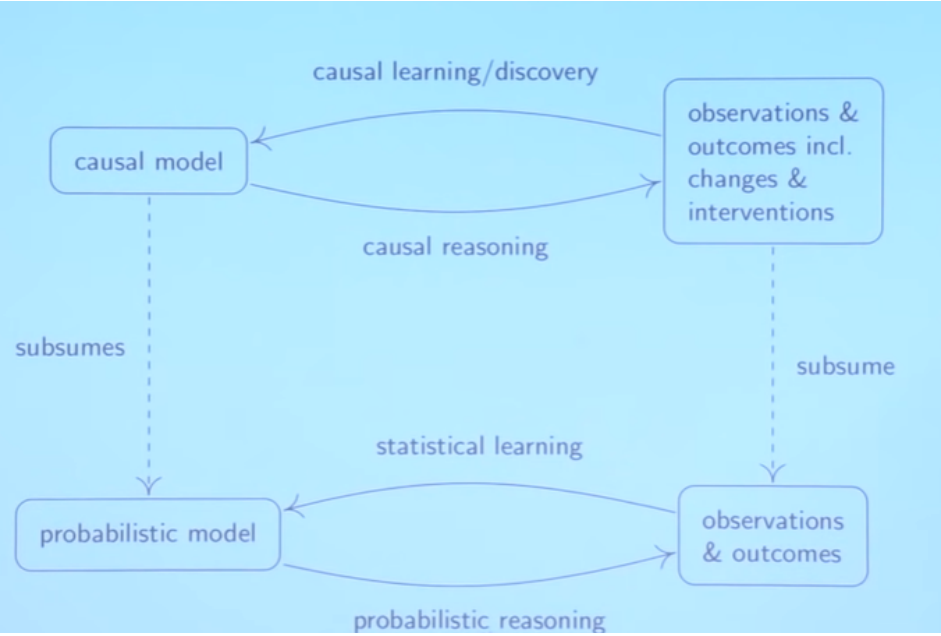

Figure: Causal reasoning

Example